H2

ANDERSON JUNIOR COLLEGE
HIGHER 2

CANDIDATE NAME

PDG

PDG INDEX NUMBER

BIOLOGY **9648/01**

Paper 1 Multiple Choice **4 September 2013**

1 hour 15 mins

Additional Materials: Multiple Choice Answer Paper **40 marks**

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name, PDG and identification number on the Answer Sheet.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

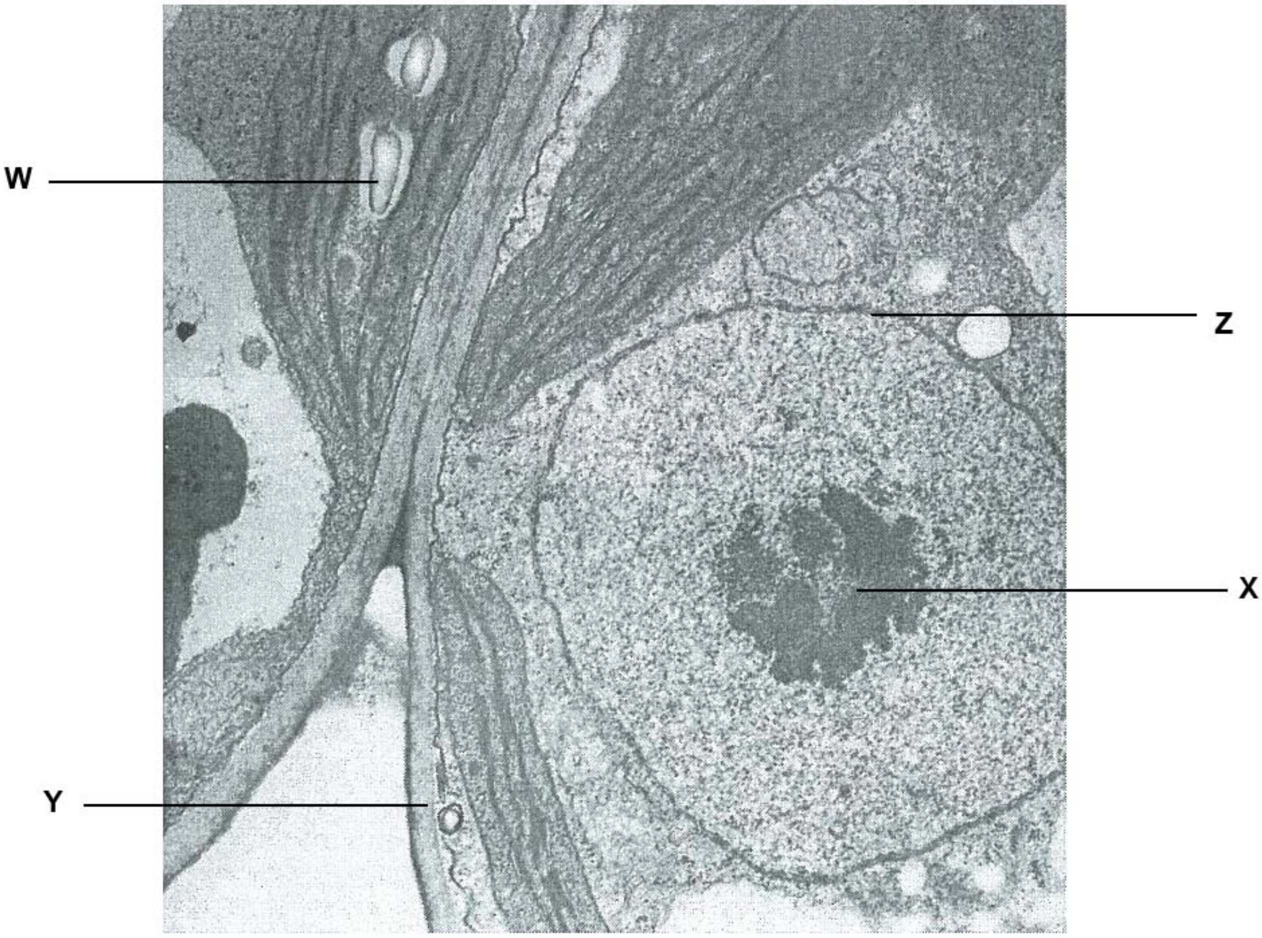
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

Calculators may be used.

1 Which row is correct for each of the components found on the cell surface membrane?

	ion pump	glycoprotein	phospholipid	cholesterol
A	a carrier protein involves in active transport	has sugar chain attached to the protein	contains 2 fatty acid chains and a phosphate head only	restrains the phospholipids movements at high temperatures
B	pumps ions against concentration gradient	is a receptor site for the signal ligand	contributes to the fluidity of the membrane	contains 4 interconnecting rings
C	use ATP	cannot move freely with the phospholipids	may be saturated or unsaturated fatty acid chains	Is a steroid
D	has a hydrophilic channel to allow ions to diffuse through	for cell-to-cell recognition	unsaturated hydrocarbon tails have kinks that allow the molecules to pack closely together	is a lipid that does not contain fatty acids

2 The electron micrograph shows two adjacent plant mesophyll cells with structures W to Z labeled.



Which of the following structure to function relationships is/ are correct?

	Structure	Function
I	W	Manufactures carbohydrates
II	X	Manufactures rRNA
III	Y	Regulates the movement of substances
IV	Z	Separates the cell from the external environment

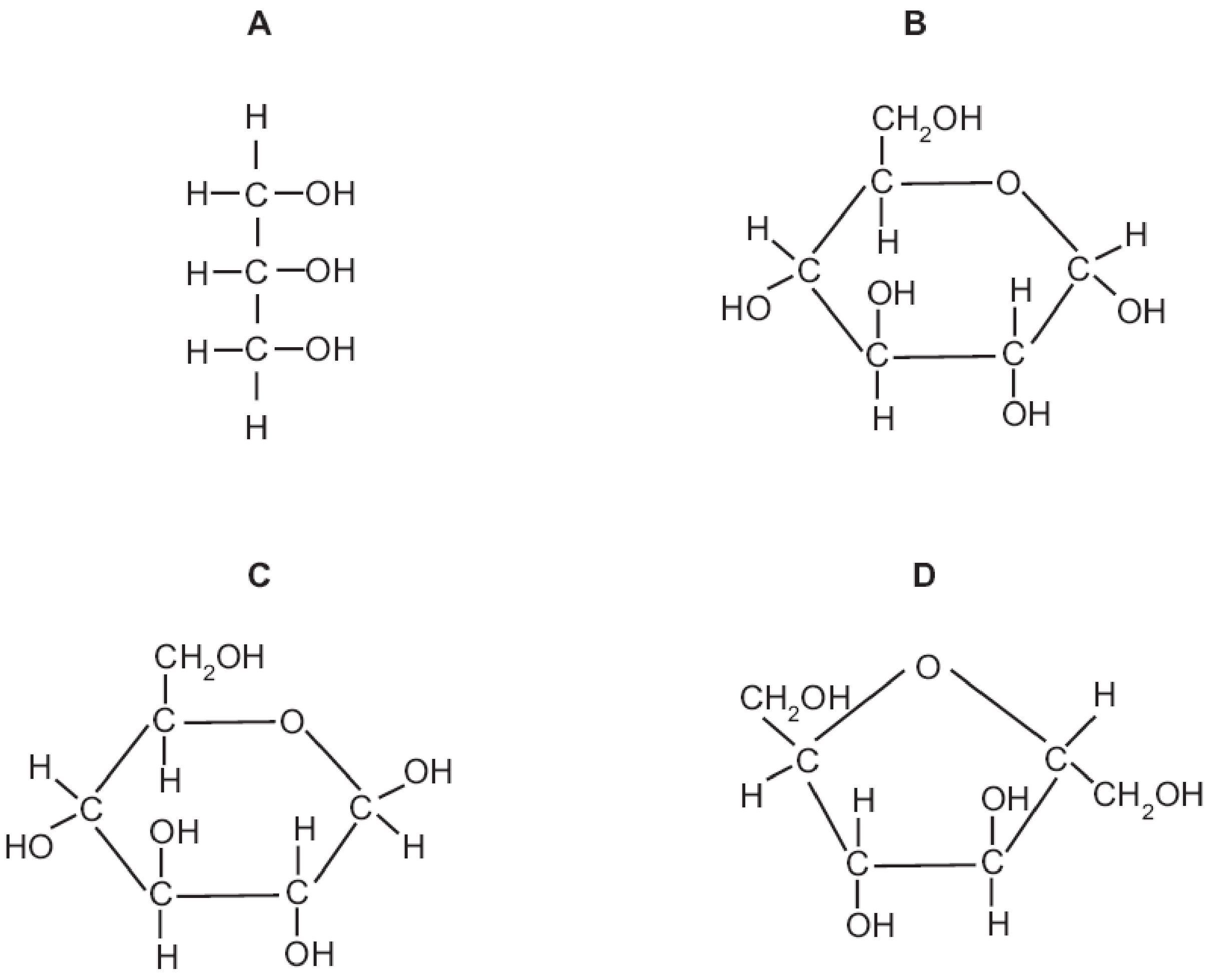
A **II** only

B **III** only

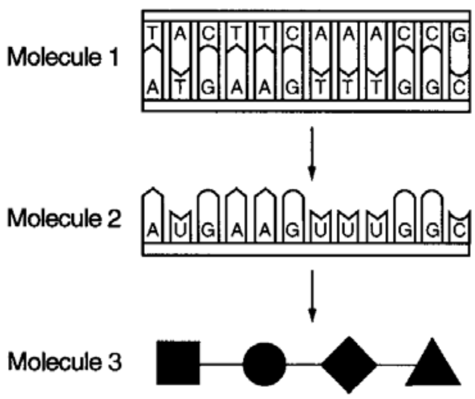
C **I** and **IV** only

D **II**, **III** and **IV** only

3 Which molecule is found in glycogen?



4 The diagram represents molecules involved in protein synthesis.



Molecules 1,2 and 3 are formed as a result of

A dehydrogenation

B condensation

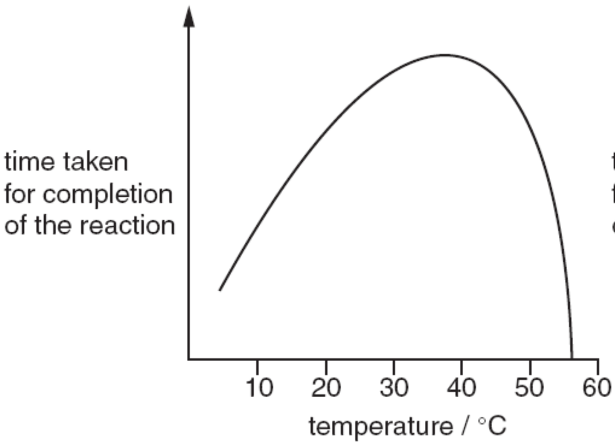
C enzymatic hydrolysis

D oxidation

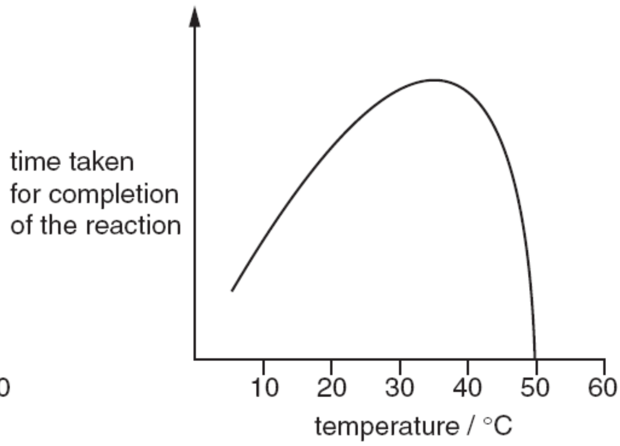
5 An enzyme is completely denatured at 50 °C. A fixed concentration of this enzyme is added to a fixed concentration of its substrate. The time taken for completion of the reaction is measured at different temperatures.

Which graph shows the results?

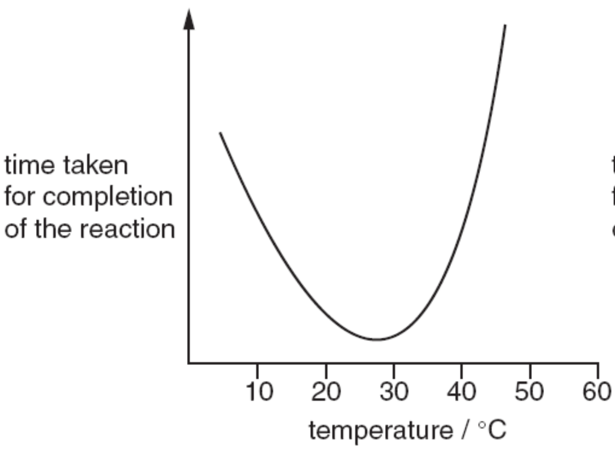
A



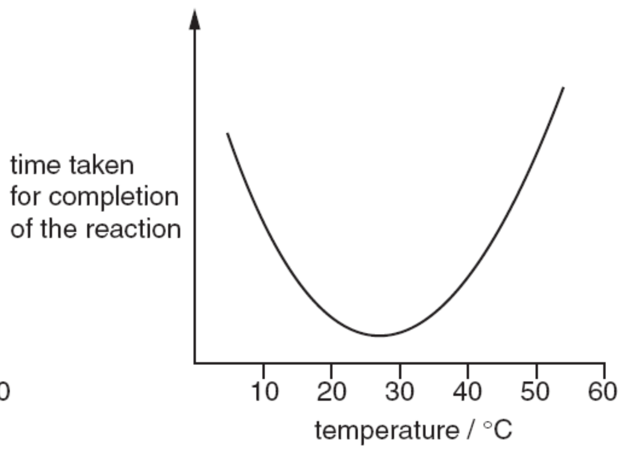
B



C



D



6 In an effort to identify allosteric inhibitors of caspases, close to 8,000 compounds were screened for their ability to bind to a possible allosteric binding site in caspase 1 and inhibit the enzyme's activity. Each compound was designed to form a disulfide bond with a cysteine near the site in order to stabilize the low-affinity interaction that is expected of an allosteric inhibitor.

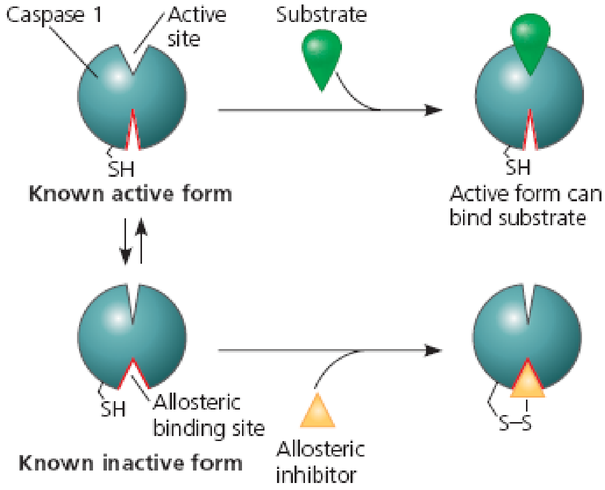


Fig. 6.1.

X-ray diffraction analysis was used to determine the structure of caspase 1 when bound to one of the inhibitors and to compare it with the active and inactive structures. The enzyme's shape when one such inhibitor was bound resembled the inactive caspase 1 more than the active form as shown in **Fig. 6.1.**

Which of the following explains how the allosteric inhibitory site on caspase 1 controls enzymatic activity?

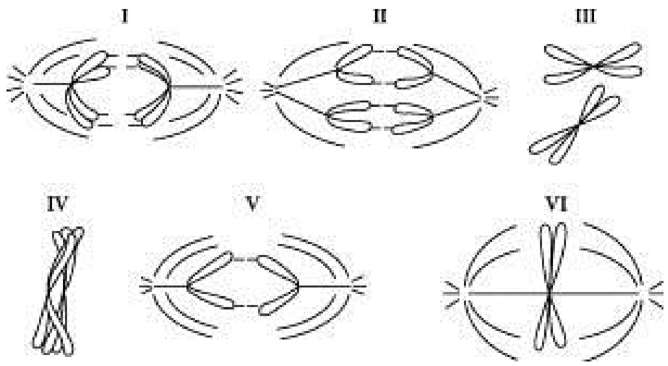
A The inhibitor locks the enzyme in its inactive form.

B The inhibitor binds reversibly at a site other than the active site.

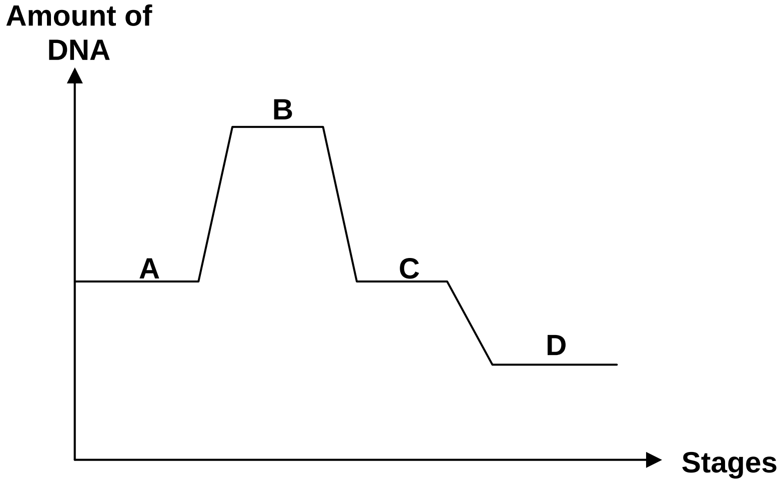
C The inhibitor binds to the active form and makes it inactive.

D The inhibitor allows caspase 1 to oscillate between catalytically active and inactive forms.

7 A cell (2n = 2) has some stages of nuclear divisions drawn which are randomly labelled I to VI.



The graph below represents **one** type of nuclear division.



Which stage corresponds to the different part of the graph correctly?

A Part A of graph – stage III

B Part B of graph – stage VI

C Part C of graph – stage V

D Part D of graph – stage IV

8 What is the basis for the difference in the synthesis of the leading and lagging strands of DNA molecules?

1 The anti-parallel arrangement of the DNA strands.

2 The RNA primers are required to initiate DNA elongation.

3 DNA polymerase joins new nucleotides to the 3' end of the growing strand.

4 Helicase and single-stranded binding proteins work at the 5'end of the DNA strand.

A 2 and 4

B 1 and 3

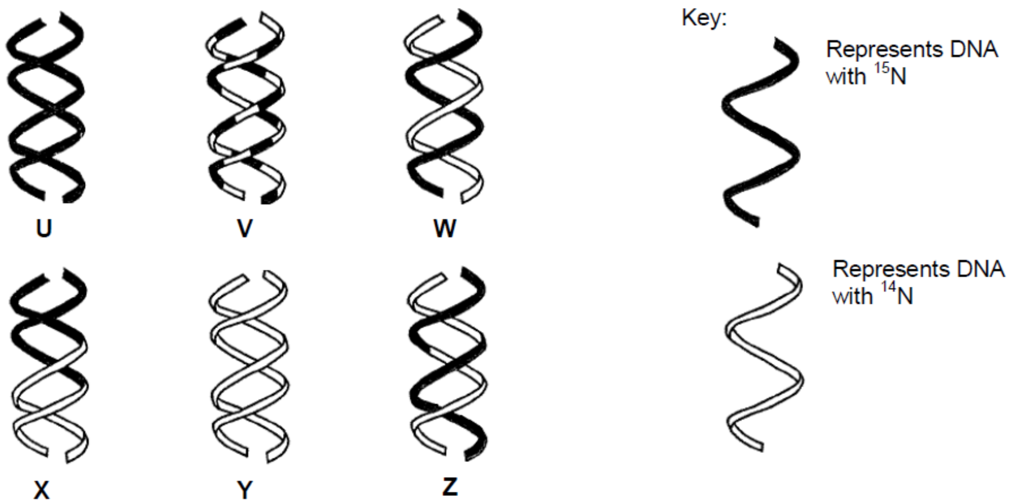
C 1 and 4

D 2 and 3

9 Three experiments were carried out to investigate the mode of DNA replication in bacteria.

- Experiment 1: Bacteria were grown for many generations with only the light isotope of nitrogen, ^{14}N, and then allowed to replicate once with the heavy isotope, ^{15}N.
- Experiment 2: Bacteria were grown for many generations with only the heavy isotope of nitrogen, ^{15}N, and then allowed to replicate once with the light isotope, ^{14}N.
- Experiment 3: Bacteria were grown for many generations with only the heavy isotope of nitrogen, ^{15}N, and then allowed to replicate twice with the light isotope, ^{14}N.

The figure shows possible DNA molecules U to Z and indicates the varying proportion of nitrogen isotopes present in each DNA molecule.



Which of the following products shows the semi-conservative mode of DNA replication?

	Experiment 1	Experiment 2	Experiment 3
A	W	W	W and Y
B	U	W	W and Y
C	W	W	X and Z
D	U	W	V and Z

10 An antibiotic, edeine, was isolated. It inhibits protein synthesis but has no effect on either DNA synthesis or RNA synthesis. When added to a translation mixture containing fully intact organelles, edeine stops haemoglobin translation after 10 seconds.

Analysis of the edeine-inhibited mixture by centrifugation showed that no polyribosomes remained by the time protein synthesis had stopped. Instead, all the mRNA accumulated, together with small ribosomal subunit and initiator tRNA.

What step in protein synthesis does edeine inhibit?

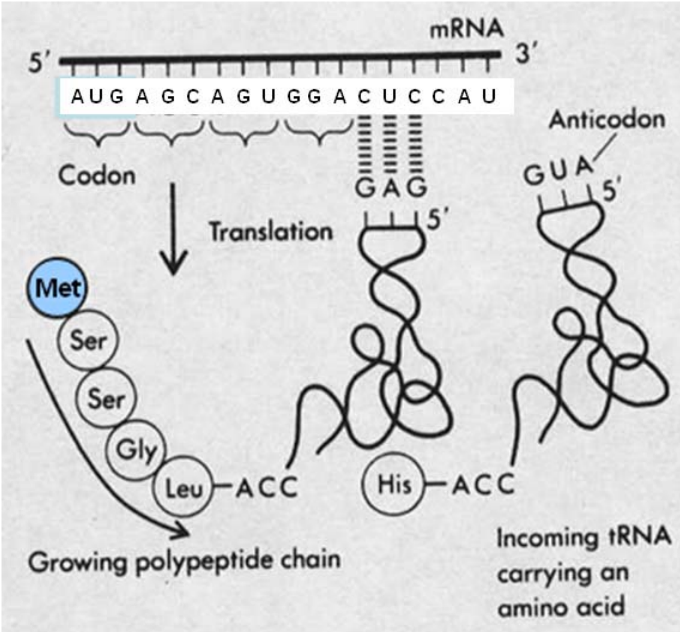
A It blocks translocation of the ribosome along the mRNA.

B It interferes with chain termination and release of peptide.

C It prevents formation of the translation initiation complex.

D It inhibits binding of amino-acyl-tRNAs to the A site of the ribosome.

11 The diagram below shows the process of translation.



Which of the following are correct?

1 The ribosome is translocating from right to left.

2 The diagram shows degeneracy of the genetic code.

3 The polypeptide does not dissociate immediately from the tRNA after histidine was added.

4 The number of hydrogen bonds formed between the respective codons and anticodons of the two tRNAs shown are equal.

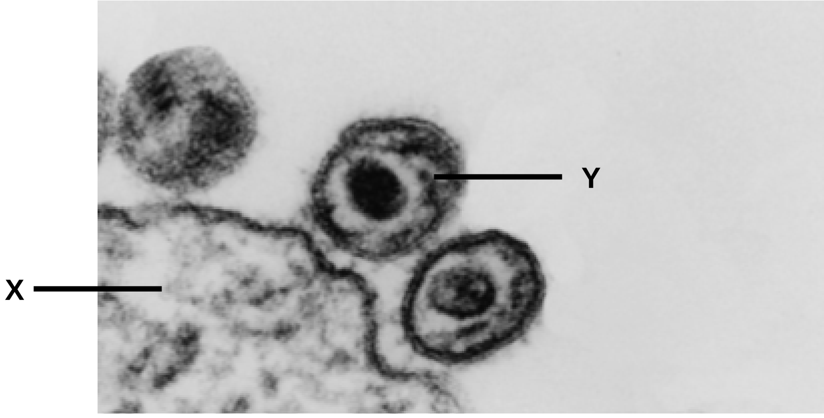
A 1 and 4

B 1 and 2

C 2 and 3

D 2 and 4

12 The diagram below shows an electron micrograph of viruses entering into a T helper cell.



Which of following statements are true?

1 Most of the components found on the plasma membrane of X can be found on the envelope of Y.

2 X and Y have the same type of nucleic acid.

3 Y enters by endocytosis.

4 Transduction occurs after Y successfully integrates its nucleic acid inside X.

A 1 only

B 1 and 3 only

C 2 and 4 only

D 3 and 4 only

13 Patients infected with the human immunodeficiency virus (HIV) tend to develop a cancer called Kaposi Sarcoma. Which is the correct order of steps to explain how HIV causes this cancer?

1 Entry of *Kaposi sarcoma* herpes virus.

2 Double-stranded provirus integrated into host cell DNA.

3 Budding of HIV kills T helper cell.

4 Development of Acquired Immunodeficiency Disease Syndrome (AIDS).

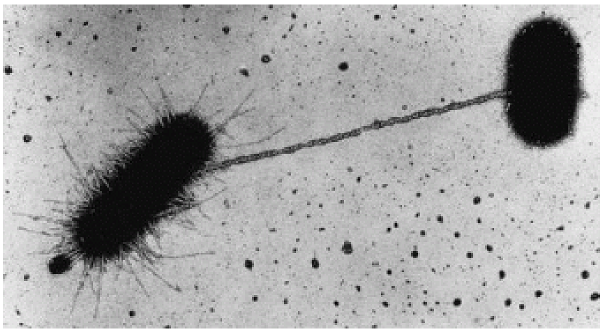
A **1 → 4 → 2 → 3**

B **2 → 3 → 1 → 4**

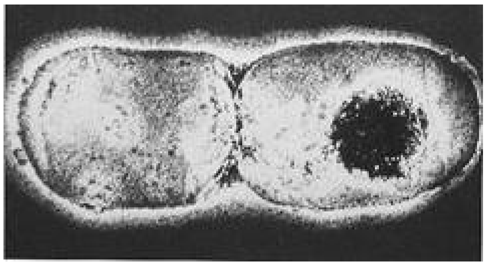
C **1 → 2 → 3 → 4**

D **1 → 2 → 4 → 3**

14 The photomicrographs below show two different processes occurring in two different species of bacteria.



Process 1



Process 2

Which of the following statements is/are true of both processes?

1 For both processes, only bacteria with genes that code for cytoplasmic bridge are involved.

2 Process 1 requires direct contact between 2 different bacteria whereas process 2 can occur with 1 bacterium.

3 Process 1 will result in an increase in the number of identical bacteria whereas process 2 will result in an increase in the number of different bacteria.

4 Both processes involve DNA replication.

A 1 and 3 only

B 2 and 4 only

C 1, 2 and 4 only

D All of the above

15 If glucose and lactose are both absent from the growth medium, and the *lac* structural genes are expressed efficiently, what would you suspect about the condition of the *E.coli* cell?

1 The cell has a mutation in the operator of the *lac* operon.

2 The cell has a mutation in the *lacI* gene.

3 The cell has a mutation in the promoter of the *lac* structural gene.

4 The cell has a mutation in the Catabolite Activator Protein (CAP) binding site.

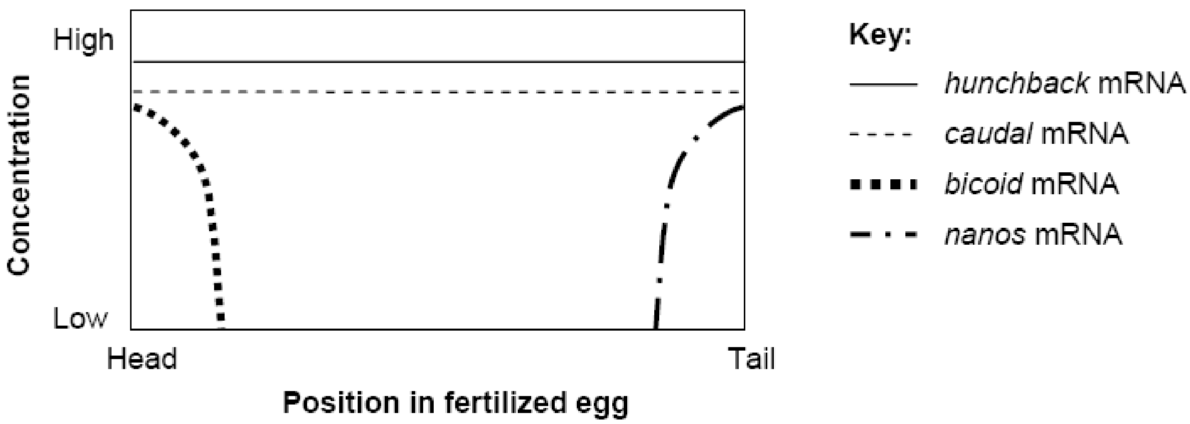
A 1 and 2 only

B 2 and 3 only

C 3 and 4 only

D 1, 2 and 4

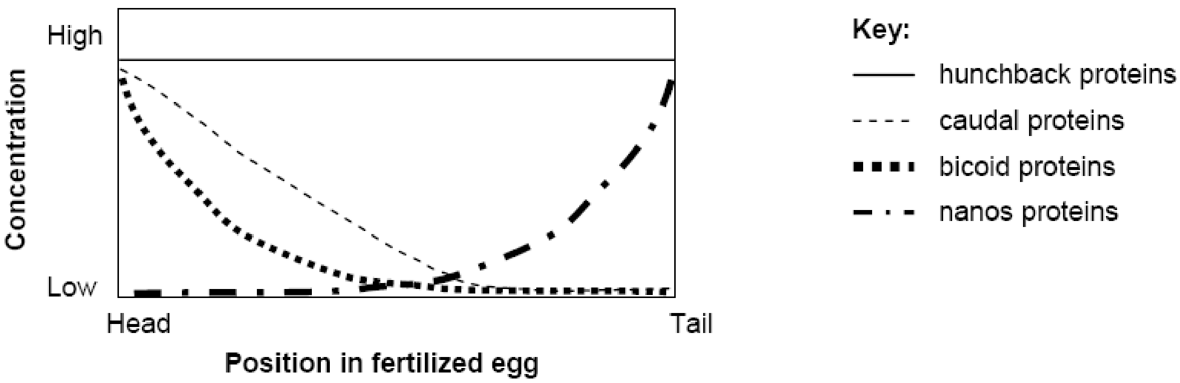
16 The figure below shows the relative distributions of four types of mRNA molecules along the head-to-tail axis in the cytoplasm of a *Drosophila* fertilized egg.



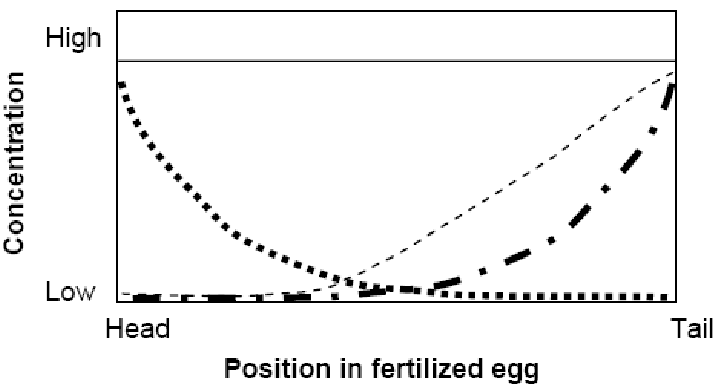
- bicoid protein binds to the 3' untranslated region of caudal mRNA to interfere with the interaction between 3' poly-A tail and 5' cap
- bicoid protein is also a specific transcription factor that binds to the enhancer of the *hunchback* gene
- nanos protein inhibits translation of hunchback mRNA

Using the information given above, determine which one of the following graphs shows the relative protein distribution along the head-to-tail axis of this fertilized egg.

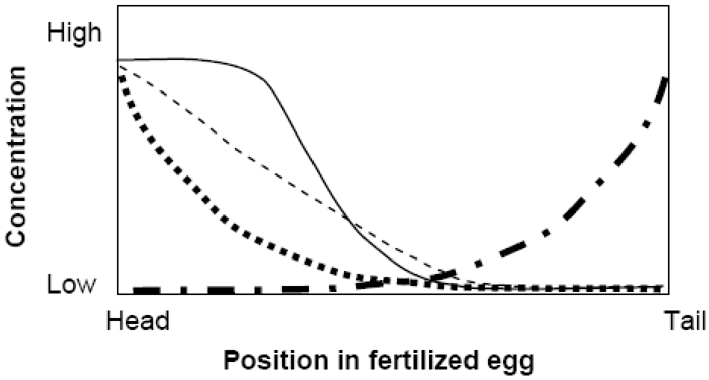
A



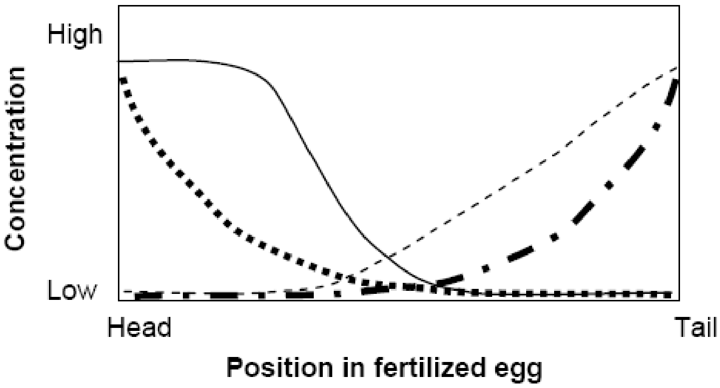
B



C



D



17 In a laboratory, the 5' 7-methylguanosine cap and the length of the polyA tail of an mRNA transcript are altered in order to examine their effects on the number of functional proteins translated.

mRNA	Alteration
1	5' cap removed with 1000-adenine tail
2	5' cap removed with 500-adenine tail
3	5'cap present with 500-adenine tail
4	5'cap present with 300-adenine tail

Using the information given in the table above, predict the relative amount of protein produced in the descending order (greatest → least).

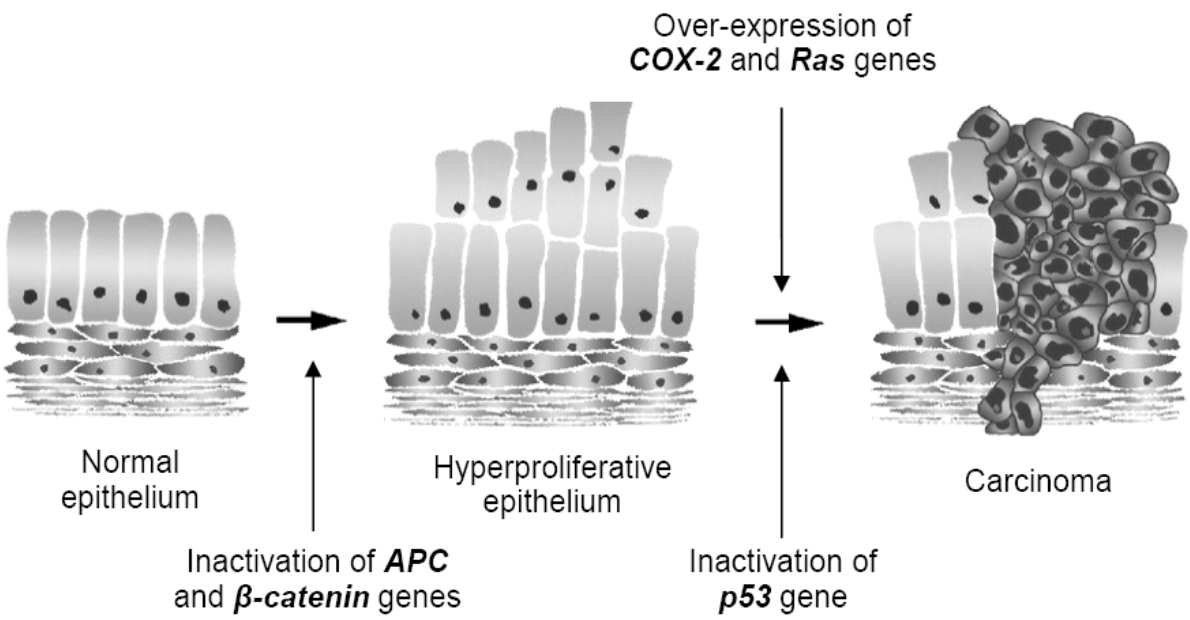
A 1,3,2,4

B 1,2,3,4

C 3,1,4,2

D 3,4,1,2

18 The diagram below illustrates the development of colorectal cancers.



Which of these statements can be inferred from this multistep model of carcinogenesis?

I Cells whose *APC* and *β-catenin* genes are inactivated have lost density dependent inhibition.

II *APC* and *β-catenin* genes are tumour suppressor genes.

III High levels of *Ras* protein are produced only when both copies of *Ras* gene are mutated.

IV Two copies of normal *p53* alleles must be present to inhibit cell division.

V Gain-of-function mutation in *COX-2* gene is a pre-requisite for the formation of carcinoma.

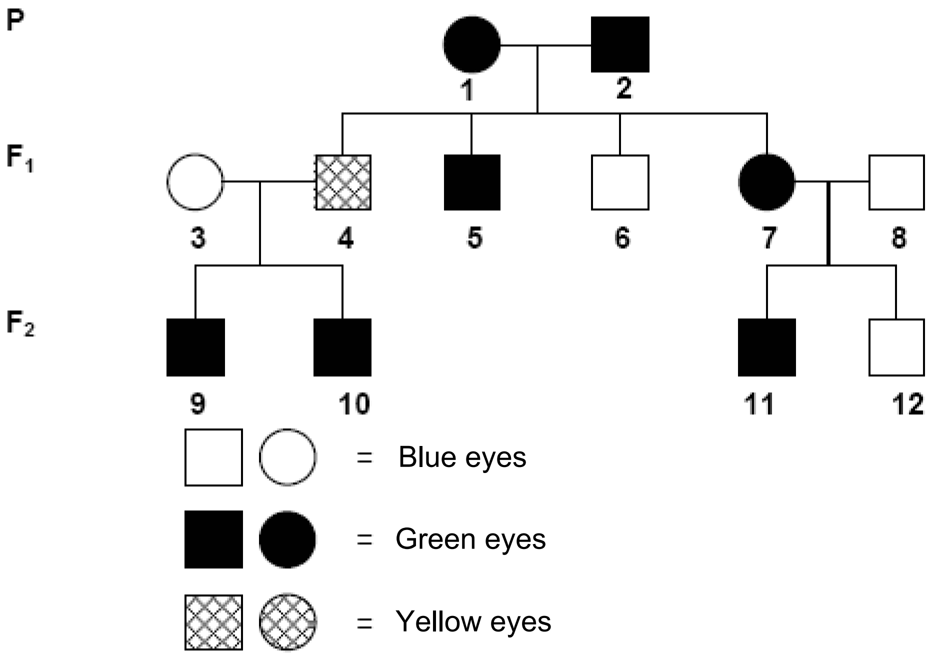
A **I**, **II** and **III**

B **II**, **III** and **IV**

C **I**, **II** and **V**

D **II, III** and **V**

19 Questions **19** and **20** refer to the following pedigree chart.



Which of the following statements is true?

A Blue eyes are dominant.

B Individual **1** is homozygous.

C Individual **4** is homozygous.

D Yellow eyes are recessive to blue eyes.

20 Which of the following statements has a 50% chance of being true?

I One of the parents of individual **2** had the same phenotype as individual **2**.

II If individual **10** married someone with green eyes, the first child would have green eyes.

III If individual **6** married a woman with green eyes, half of the offspring would have blue eyes.

A **II** only

B **I** and **III** only

C **II** and **III** only

D All of the above

21 In Shorthorn cattle, the allele for hornless is dominant to the allele for the presence of horns. Coat colour can be red (genotype C^RC^R), roan (genotype C^RC^W) or white (genotype C^WC^W). A roan bull, heterozygous for the hornless trait, is crossed with a cow of the same genotype.

What is the probability that a calf from this cross would have the same genotype as its parents?

A 1/4

B 3/8

C 1/2

D 3/4

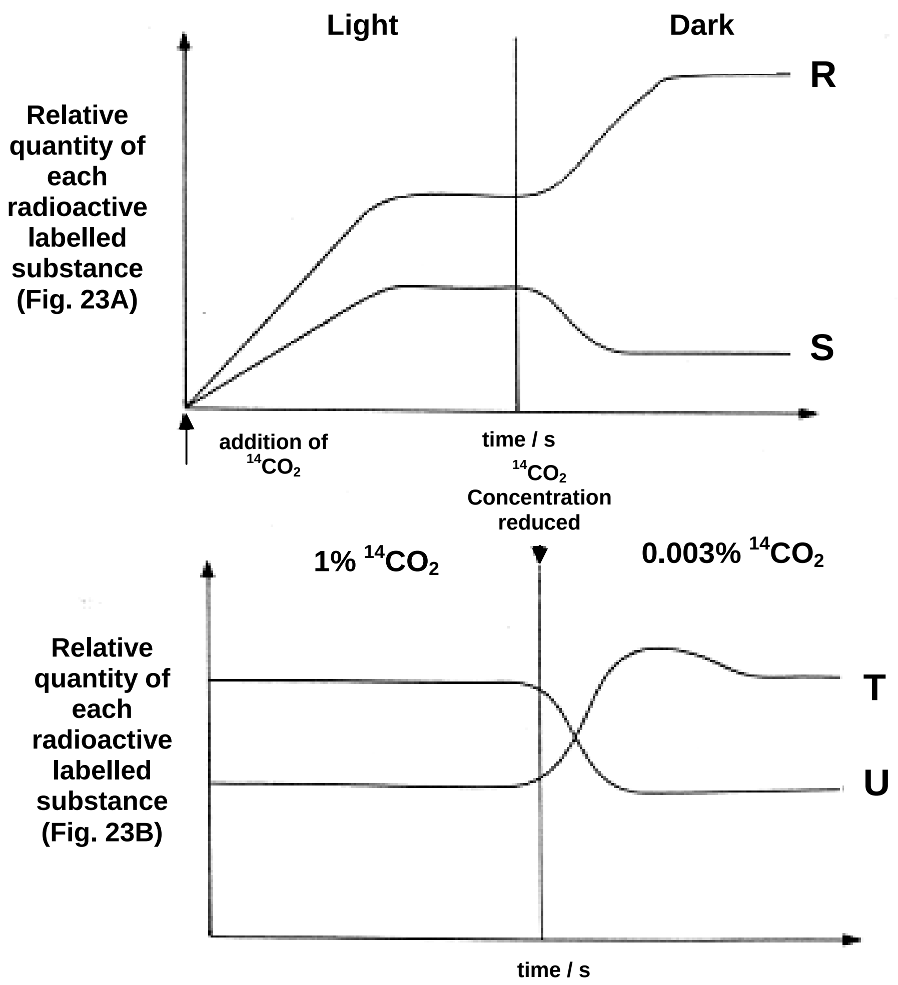
22 A yellow flower, green-stemmed plant with the genotype **YYrr** was crossed with a white flower, red-stemmed plant with the genotype **yyRR**. The F1 plants were allowed to self fertilise, A χ^2 test was carried out on the results obtained for the F2 generation. Part of the table of values for χ^2 is shown.

degrees of freedom	p = 0.5	p = 0.1	p = 0.05	p = 0.01	p = 0.001
1	0.46	2.71	3.84	6.64	10.83
2	1.39	4.6	5.99	9.21	13.82
3	2.37	6.25	7.82	11.34	16.27
4	3.37	7.78	9.49	13.28	18.46
5	4.35	9.24	11.07	15.09	20.52

The value of χ^2 in this investigation was 10.6.
What is the probability of this value of χ^2 and do the results fit the expected ratio?

	probability	results fit expected ratio
A	between 0.01 and 0.05	No
B	between 0.01 and 0.05	Yes
C	between 0.05 and 0.1	Yes
D	between 0.1 and 0.5	No

23 Fig. 23A and Fig. 23B show the results of two separate experiments on carbon dioxide fixation in photosynthesis using a unicellular green alga and the radioactive isotope ^{14}C to label carbon dioxide. The algae were actively photosynthesizing before the start of both experiments.



With reference to Fig. 23A and 23B.identify the graphs **R, S, T,** and **U.**

	R	S	T	U
A	RuBP	GP/PGA	GP/PGA	RuBP
B	GP/PGA	RuBP	GP/PGA	RuBP
C	GP/PGA	RuBP	RuBP	GP/PGA
D	RuBP	GP/PGA	RuBP	GP/PGA

24 Which of the following statements is true regarding cyclic and non-cyclic photophosphorylation?

A Cyclic photophosphorylation only requires light at 680nm to function optimally while non-cyclic photophosphorylation requires light at both 680nm and 700nm to function optimally.

B Cyclic photophosphorylation is dependent on products generated through non-cyclic photophosphorylation.

C The electrochemical gradient generated to drive ATP production by the non-cyclic is different from that of the cyclic photophosphorylation.

D Only cyclic photophosphorylation can function in the absence of photosystem II.

25 When cyanide is bound to cytochrome oxidase, the cell can no longer produce ATP aerobically. Which statement below best explains this?

A It prevents reduced NADP from being oxidised to NADP, hence preventing electron flow down the electron transport chain.

B It prevents oxygen from accepting electrons and protons to form water, hence preventing electron flow down the electron transport chain.

C It prevents photolysis from occurring to produce oxygen.

D It prevents pyruvate from being synthesised in glycolysis, hence stopping the Krebs cycle.

26 Which of the following shows correctly the differences in the products of glycolysis and Krebs cycle per glucose molecule?

	Glycolysis	***Krebs cycle***
A	4ATP, 1 reduced NAD	4 reduced NAD, 1 reduced FAD, 1 ATP
B	4ATP, 2 reduced NAD	8 reduced NAD, 2reduced FAD, 2 ATP
C	2 net ATP, 1 reduced NAD	3 reduced NAD, 1 reduced FAD, 1 ATP
D	2 net ATP, 2 reduced NAD	6 reduced NAD, 2 reduced FAD, 2 ATP

27 Certain drugs act at synapses and affect the action of neurotransmitter substances. The table shows the effects of four different drugs.

Drug	*Effect*
1	Inhibits the enzyme cholinesterase
2	Prevents the release of acetylcholine
3	Opening of Ca^{2+} gates on the pre-synaptic membrane.
4	Binds to the post-synaptic membrane receptors.

Which of the two drugs would most likely to *encourage* the response from a skeletal response to an electrical impulse arriving at the neuro-muscular junction?

A 1 and 3

B 1 and 4

C 2 and 3

D 2 and 4

28 What is the role of active transport in the transmission of nerve impulses by neurones?

A Propagates an action potential by pumping sodium ions across the membrane out of the neurone.

B Propagates an action potential by pumping sodium ions across the membrane into the neurone.

C Initiates the action potential needed for the transmission of an impulse by pumping calcium ions out of the endoplasmic reticulum.

D Establishes the resting potential needed for the transmission of an impulse by pumping sodium and potassium ions across the membrane.

29 Three proteins isolated from a human cell were investigated for their involvement in cell signalling pathways.

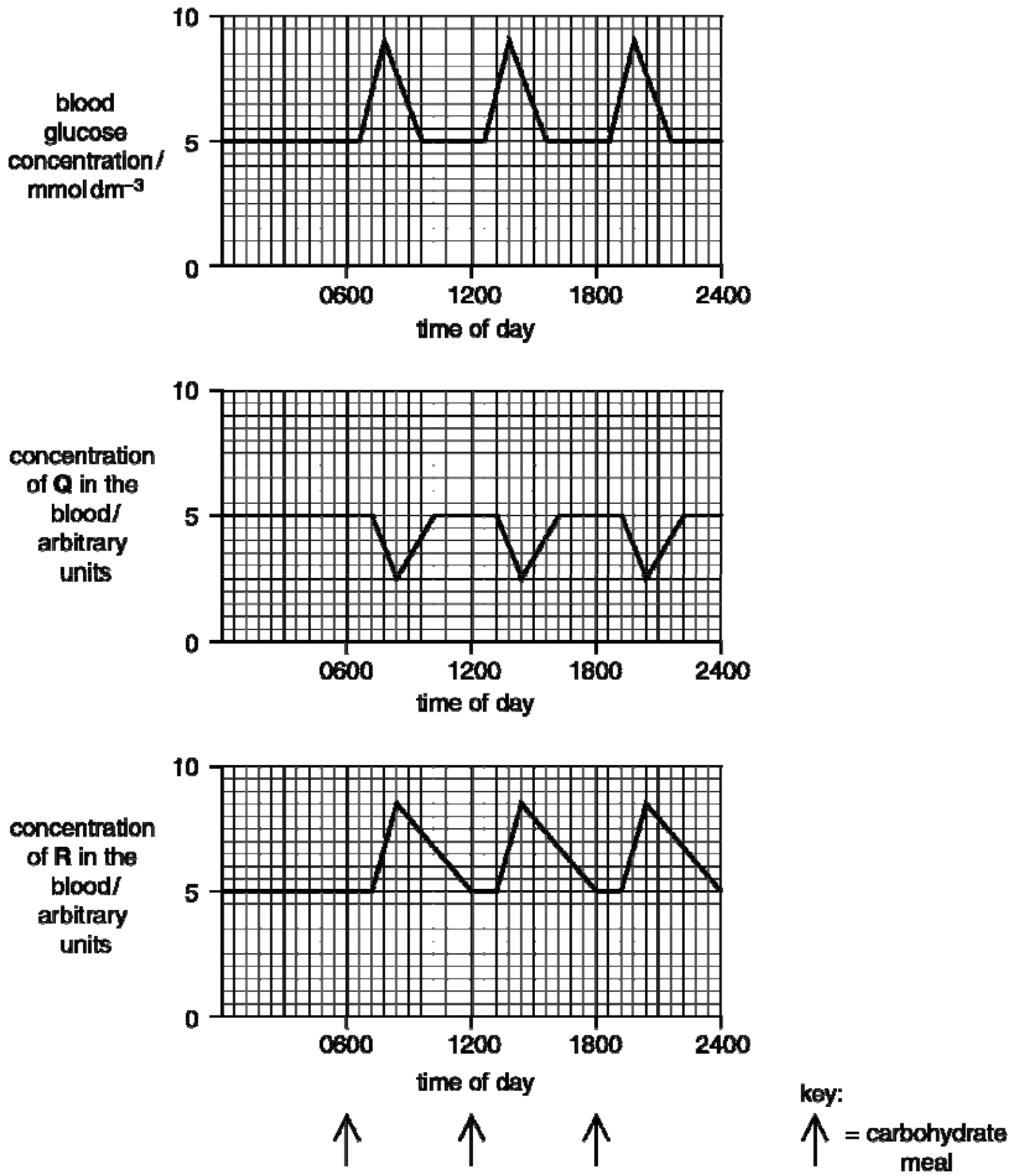
	Protein A	Protein B	Protein C
ATPase activity	+	-	-
GTPase activity	-	-	+
Kinase activity	-	+	-
Trans-membrane Domain	+	+	+

Key: (+) = present, (-) = absent

Which of the following correctly shows the identity of these three proteins?

	Protein A	**Protein B**	**Protein C**
A	Na^+ - K^+ Pumps	Insulin receptor	Glucagon Receptor
B	Glucagon Receptor	Insulin receptor	Na^+ - K^+ Pump
C	Na^+ - K^+ Pump	Glucagon Receptor	Insulin receptor
D	Insulin receptor	Glucagon Receptor	Na^+ - K^+ Pump

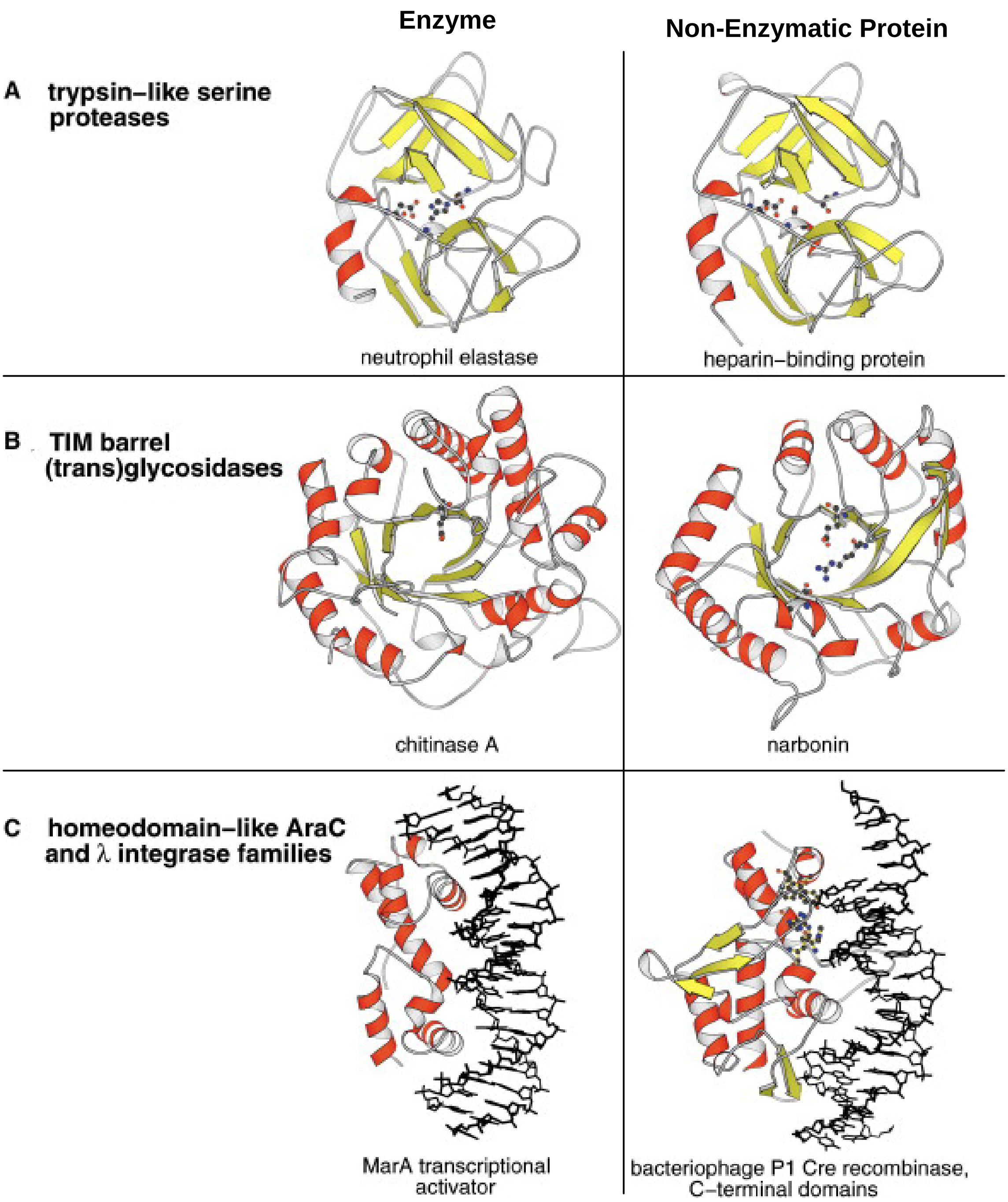
30 The figure shows the relationship between glucose concentration in the blood and the concentrations in the blood of the two hormones, **Q** and **R**.



Which hormones (Q and R) promote the processes shown?

	conversion of glycogen to glucose in liver cells	respiration of glucose in liver cells	uptake of glucose by muscle cells
A	Q	R	Q
B	Q	R	R
C	R	Q	Q
D	R	Q	R

31 Three pairs of enzymes and non-enzymatic proteins with similar primary structures are shown.



Which of the following correctly explains the type of structures for each pair?

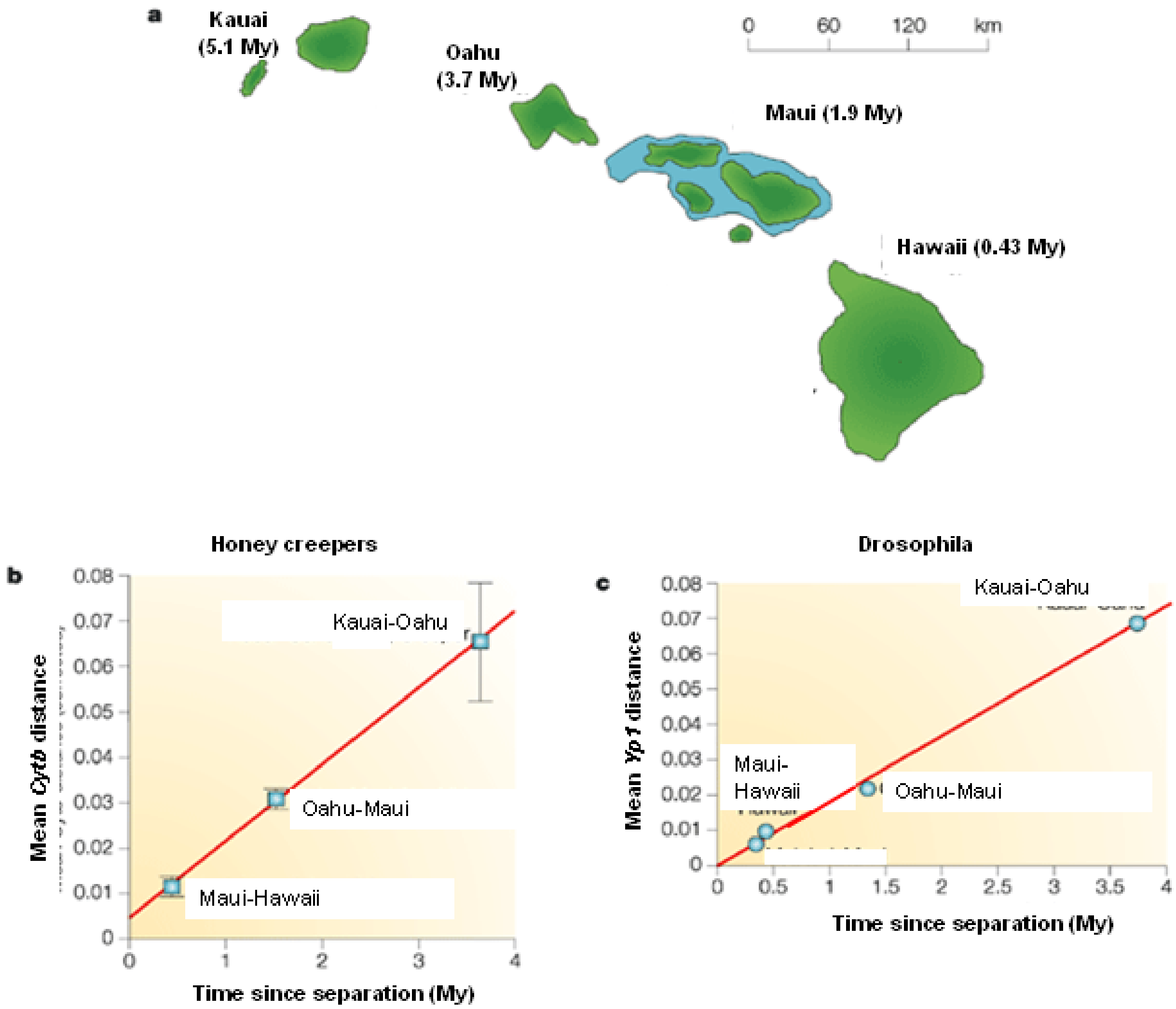
A Analogous structures as they perform similar functions.

B Analogous structures as their tertiary structures are similar.

C Homologous structures as they perform different functions.

D Homologous structures as they are all polypeptides folded into a 3D configuration.

32 The volcanic islands that were formed millions of years ago, range from Kauai (the oldest) to Hawaii (the youngest). *Cytb* gene from honey creepers and *Yp1* gene from *Drosophila* were analysed for divergence.



Which of the following statement is **wrong**?

A Geographical isolation prevented colonization of newly formed islands.

B There is a positive linear correlation between genetic distance and island age.

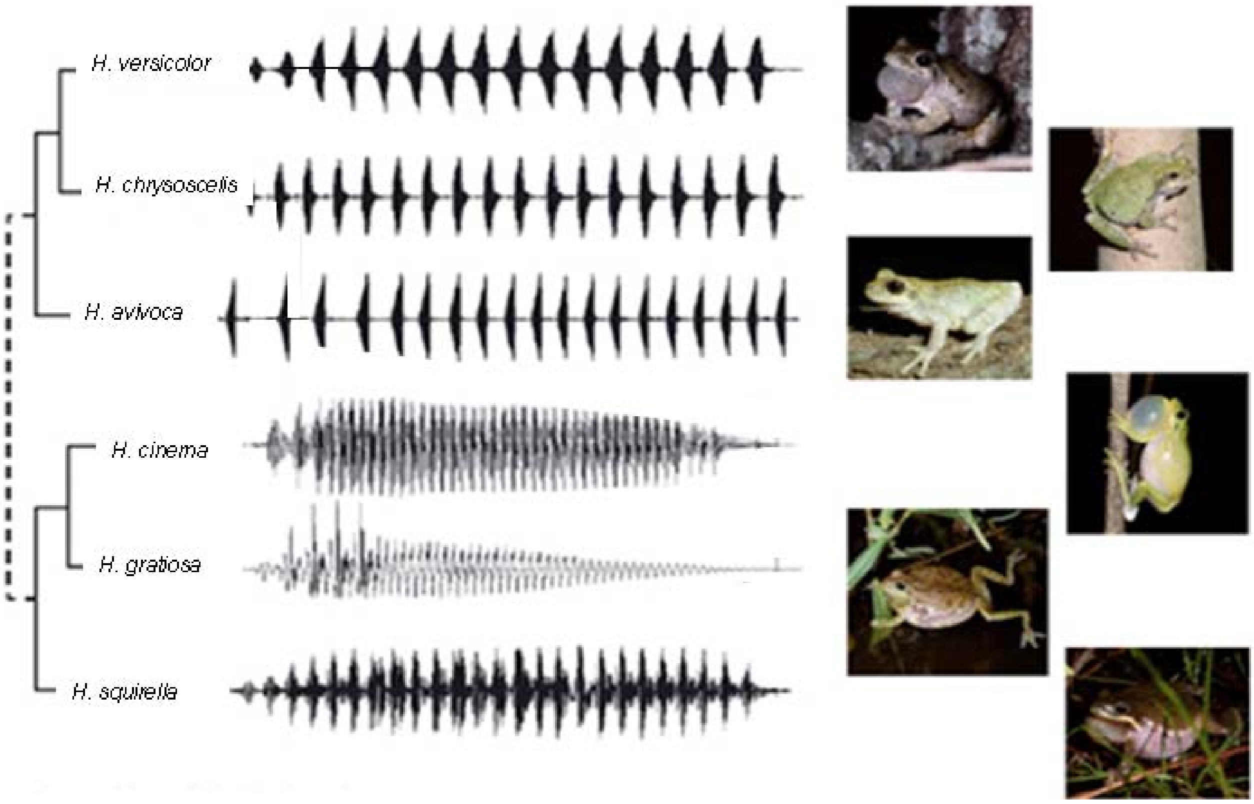
C *Cytb* gene and *Yp1* gene are chosen because they are essential genes.

D Genetic drift is a factor that contributes to the increase in the mean genetic distance.

33 Which of the following describes the concept of gene flow correctly?

	Plays a role in speciation	Explains why population is the smallest unit to evolve	Affects the strength of selection pressure
A	No	No	Yes
B	No	No	No
C	Yes	Yes	Yes
D	Yes	Yes	No

34 The calls of six different species of frogs belonging to the *Hyla* genus are recorded and shown.



Which of the following can be inferred from the chart?

1 Frogs with more similar call patterns are more closely related.

2 A frog species can be identified by looking at the duration, intensity, and frequency of the call.

3 The call of each species of frog acts as a selection pressure.

4 These calls are a form of isolation mechanism.

A 1 and 2 only

B 1, 2 and 4 only

C 1 and 3 only

D All of the above

35 Which primers should be paired to amplify, by PCR, the DNA flanked by the two indicated sequences?

5' – GGAATTCGT -- // -- AATGCTACC – 3'

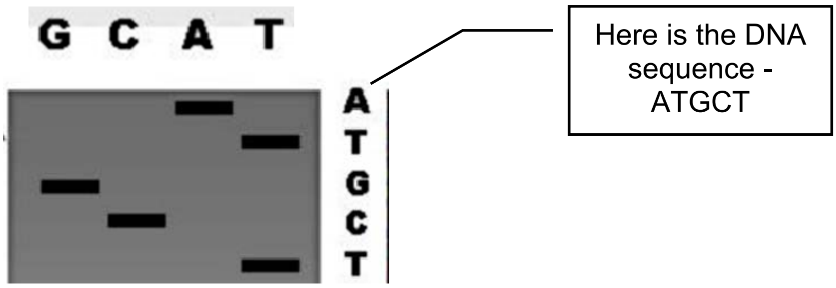
A 5' ACGAATTCC 3' & 5' AATCGTACC 3'

B 5' AATGCTACC 3' & 5' GGAATTCGT 3'

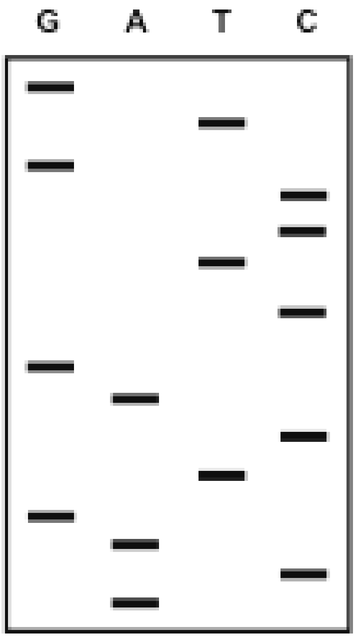
C 5' GGTAGCATT 3' & 5' GGAATTCGT 3'

D 5' ACGAATTCC 3' & 5' GGTAGCAAT 3'

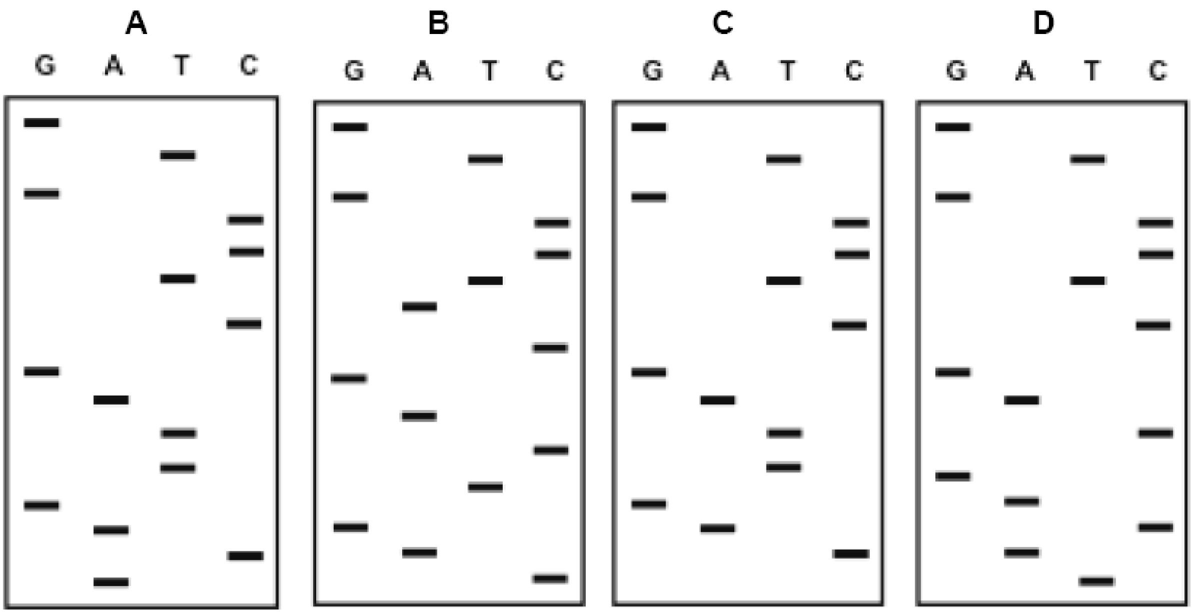
36 In people with a gene mutation where a base insertion has occurred, the protein formed has a very different primary structure. To identify the presence of this mutant allele, DNA nucleotide sequences were compared using gel electrophoresis. Read from the top to the bottom for the DNA sequence. Here's an example:



The electrophoresis result from the DNA of a normal allele of the gene is shown below.



Which diagram represents the DNA sequence for the mutant allele of this gene?

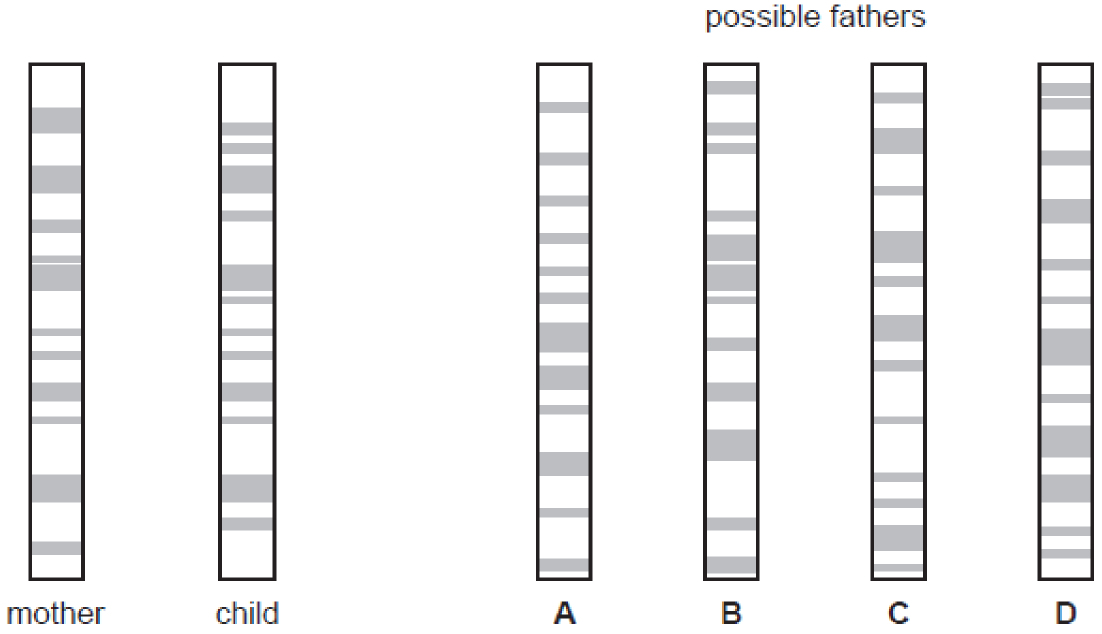


37 Genetic profiling can be used to determine the paternity of a child.

DNA from the mother and the child is cut into fragments, separated by electrophoresis and made visual using a stain.

The diagram shows the genetic profiles of a mother and child, and four possible fathers.

Who is the father? B



38 Which of the following statements are **TRUE** about all stem cells?

1 Stem cells can be induced to differentiate by environmental signals.
2 Stem cells are easily isolated and propagated.
3 Stem cells are able to develop into whole organisms if implanted into the womb.
4 Stem cells make more stem cells under appropriate conditions.

A 1 and 4
B 2 and 3
C 1, 3 and 4
D 1, 2, 3 and 4

39 Somaclonal variation may arise during *in vitro* culturing of plant tissues because

A crossing over occurs between homologous chromosomes during meiosis in the cultured plant cells, leading to genetic recombination.

B some cells are committed to differentiation into specialized plant tissues, therefore may silence the expression of certain genes.

C of increased rate of mutation due to high growth rate of cells which is induced by the large amount of growth regulators used.

D pollinating insects inserts DNA sequences containing genes from other plant cells into the cultured cells' genome, leading to genetic variation.

40 Corn is a major crop grown in Europe. In the past, it was either ruined by attack from the corn borer larvae or intensively sprayed with pesticides each year. The biotech company Novartis gained approval to insert a *Bt* gene into corn. This gene codes for a protein that kills the larvae feeding on the corn. The genetically engineered *Bt* corn initially thrived without the addition of any pesticides, but later suffered damage from pests again.

Which of the following are possible reasons why the *Bt* corn is again under attack from pests?

I	A strain of resistant larvae has emerged as a chance mutation. The frequency of the gene conferring resistance as well as the number of larvae with resistance has increased. The *Bt* gene is now ineffective.
II	Corn is being attacked by other pests which are not killed by the *Bt* gene protein.
III	The *Bt* allele has undergone insertional mutagenesis. This mutagenesis has had a selective advantage in seed fertility.

A **I** and **II** only

B **I** and **III** only

C **II** and **III** only

D All of the above

Paper 1 Answer scheme

Qns	Ans	Remarks	Qns	Ans	Remarks
1	B		21	A	
2	A		22	A	
3	B		23	D	
4	B		24	D	
5	C		25	B	
6	A		26	D	
7	C		27	C	
8	B		28	D	
9	A		29	A	
10	C		30	B	
11	C		31	C	
12	A		32	A	
13	B		33	D	
14	B		34	B	
15	A		35	C	
16	D		36	B	
17	D		37	B	
18	C		38	A	
19	C		39	C	
20	A		40	D	